## 3.6 Localisation

Algebraically, localisation can be seen as enlargening a ring to include inverses. In terms of the ideal structure this means removing (proper) ideals which contain the newly inverted elements. Geometrically ideals correspond to points/subsets, so localisation may be viewed as reducing the set of interest.

Recall the definition of multiplicative set. Some rather canonical examples are as follows

**Example 3.6.1**
*The set $S_f = \{1, f, f^2, ...\}$ is m.c. but not necessarily saturated. As an example consider $A = \mathbb{Z}$ and $S_n = \{1, n, n^2, ...\}$ for $n$ composite. Then $pq \in S_n$ but $p \notin S_n$.*

*We denote the localisation of $A$ at $S_f$ by $A_f$ (or alternatively $A[1/f]$).*

**Example 3.6.2**
*If $\mathfrak{p} \lhd A$ is a prime ideal, then $A \setminus \mathfrak{p}$ is a saturated multiplicative set. More generally, we show later that $S$ is a saturated multiplicative set if and only if it's of the form*

$$A \setminus \bigcup_i \mathfrak{p}_i$$

*for some family of prime ideals. We denote the localisation of $A$ at $\mathfrak{p}$ by $A_{\mathfrak{p}}$.*

### 3.6.1 Rings

**Definition 3.6.3 (Localisation of a ring)**
*Let $A$ be a ring and $S$ a multiplicative set. Define the set*

$$S^{-1}A = \left\{ \frac{a}{s} \;\middle|\; a \in A \, s \in S \right\}$$

*under the equivalence relation*

$$\frac{a}{s} = \frac{b}{t} \iff u(at - bs) = 0 \quad \text{some } u \in S.$$

*then this is a ring in the obvious way*

**Proposition 3.6.4**
*The set $S^{-1}A$ is a ring under the obvious ring operations. It is non-zero precisely when $S$ is proper. There is a canonical homomorphism*

$$i_S : A \to S^{-1}A$$
$$a \to \left[\frac{a}{1}\right]$$

*a)* $\ker(i_S) = \mathrm{Ann}(S)$
*b)* $S^{-1}A$ *is the zero-ring if and only if* $0 \in S$ *if and only if there exists* $s, t \in S$ *such that* $st = 0$.
*c)* $i_S(s)$ *is invertible for all* $s \in s$
*d)* $i_S$ *is injective if and only if* $S$ *has no zero-divisors*
*e) This is an isomorphism if and only if* $S \subseteq A^\star$ *already consists only of invertible elements (e.g.* $S = \{1\}$*).*

*Proof.*
a) This follows by the definitions
b) $\frac{1}{1} = \frac{0}{0} \iff s = 0$ for some $s \in s$ by the definitions
c) $\frac{s}{1} \cdot \frac{1}{s} = \frac{s}{s} = \frac{1}{1}$
d) This follows from the first part.
e) If $S \subseteq A^\star$ then it contains no zero-divisors and $i_S$ is injective. Further it's clear that $\frac{a}{s} = \frac{as^{-1}}{1}$ so that the map is surjective. Similarly if the map is bijective $S$ does not contain zero-divisors and $\frac{1}{s}$ is in the image. Therefore there is $a$ such that $tas = 1$ for some $t$, which implies $s$ is invertible.

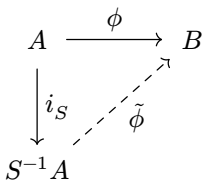
■

Note when $A$ is an integral domain and $S$ is proper then the equivalence relation may be weakened to $at - bs = 0$.

**Proposition 3.6.5 (Universal Property)**
*Let $\phi : A \to B$ be a ring homomorphism and $S$ a multiplicative set. Then*

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*a) There is a unique morphism $\tilde{\phi}$ making the diagram commute*




*if and only if $\phi(S) \subseteq B^\star$. In this case it's given by*

$$\tilde{\phi}\left(\frac{a}{s}\right) = \phi(a)\phi(s)^{-1}$$

*b)* $\ker(\tilde{\phi}) = S^{-1}\ker(\phi)$

*Proof.*

a) If $\tilde{\phi}$ exists then $1 = \tilde{\phi}(1) = \tilde{\phi}(\frac{s}{1}\cdot\frac{1}{s}) = \tilde{\phi}(\frac{s}{1})\tilde{\phi}(\frac{1}{s}) = \phi(s)\tilde{\phi}(\frac{1}{s})$. Which shows that $\phi(S) \subseteq B^\star$ and $\phi(s)^{-1} = \tilde{\phi}(\frac{1}{s})$.
Conversely suppose $\phi(S) \subseteq B^\star$ then we claim that the given mapping is well-defined. For

$$\frac{a}{s} = \frac{a'}{s'} \implies s''(s'a - sa') = 0 \implies \phi(s'')\phi(s')\phi(a) = \phi(s'')\phi(s)\phi(a')$$

Multiply by the appropriate inverses to find

$$\phi(a)\phi(s)^{-1} = \phi(a')\phi(s')^{-1}$$

It's clearly a multiplicative homomorphism. Further it's additive because

$$\begin{aligned}\tilde{\phi}\left(\frac{a}{s} + \frac{b}{t}\right) &= \tilde{\phi}\left(\frac{at+bs}{st}\right)\\ &= \phi(at+bs)\phi(st)^{-1}\\ &= \phi(a)\phi(t)\phi(s)^{-1}\phi(t)^{-1} + \phi(b)\phi(s)\phi(s)^{-1}\phi(t)^{-1}\\ &= \phi(a)\phi(s)^{-1} + \phi(b)\phi(t)^{-1}\\ &= \tilde{\phi}\left(\frac{a}{s}\right) + \tilde{\phi}\left(\frac{b}{t}\right)\end{aligned}$$

b) Suppose $\tilde{\phi}(\frac{a}{s}) = 0$ then clearly $a \in \ker(\phi) \implies \frac{a}{s} \in S^{-1}\ker(\phi)$. The converse is clear.

■

In the case that $A$ is an integral domain then generally everything becomes a lot simpler.

**Example 3.6.6 (Quotient Field)**
*Let $A$ be an integral domain then $A \setminus 0 = A^\star$ and we define the* ***quotient field***

$$\mathrm{Frac}(A) \coloneqq (A \setminus 0)^{-1}A$$

**Proposition 3.6.7 (Quotient Field contains all localisation)**
*Let $A$ be an integral domain, and* $\mathrm{Frac}(A)$ *the quotient field. Define another model for $S^{-1}A$ as follows*

$$S^{-1}A \coloneqq \left\{\frac{a}{s} \in \mathrm{Frac}(A) \;\middle|\; a \in A\, s \in S\right\}$$

*The canonical map $A \to S^{-1}A \subset \mathrm{Frac}(A)$ is injective, and satisfies the universal property for localisation.*

*Proof.* It's injective because $A$ has no zero-divisors. That it satisfies the universal property is very similar as before.■

**Proposition 3.6.8 (Directed Limit)**
*Let $S_i$ be a family of multiplicatively closed sets directed by inclusion, such that $S = \bigcup_i S_i$ is multiplicatively closed. Then there is a canonical isomorphism*

$$\varinjlim_i S_i^{-1}A \to S^{-1}A$$

*induced by the canonical maps*

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$$S_i^{-1}A \to S^{-1}A$$

*Proof.* The canonical maps $i_{S_iS}$ induce a unique morphism

$$\varinjlim_i S_i^{-1}A \to S^{-1}A$$
$$[a_i/s_i] \to a_i/s_i$$

by the universal property. An element on the right hand side is written $a/s$ for some $s \in S$. By hypothesis $s \in S_i$ for some $i$, therefore it is surjective. Suppose we have two elements $[a_i/s_i]$ and $[a_j/s_j]$ on the left hand side which become equal in $S^{-1}A$. Then by definition $s_k(s_ja_i - a_js_i) = 0$ for some $s_k \in S_k$. Since it's a directed system we can find $S_l$ containing $S_i, S_j, S_k$. Then by definition $a_i/s_i = a_j/s_j$ in $S_l^{-1}A$ and we see that $[a_i/s_i] = [a_j/s_j]$. Therefore the given morphism is also injective as required. ∎

**Lemma 3.6.9 (Localisation Respects Contraction of Multiplicative Sets)**
*Let $A \subseteq B$ be subrings of a common ring $C$, and $T \subset B$ a multiplicative set. Writing $S := T \cap A$ (a multiplicative set of $A$), and realising both localisations as subrings of $C$ (or of a common ring of fractions), $S^{-1}A \subseteq T^{-1}B$.*

*In particular, for $\mathfrak{q}$ a prime ideal of $B$ and $\mathfrak{p} := \mathfrak{q} \cap A$, taking $T := B \setminus \mathfrak{q}$ (so $S = A \setminus \mathfrak{p}$) gives $A_\mathfrak{p} \subseteq B_\mathfrak{q}$.*

*Proof.* For $a/s \in S^{-1}A$ ($a \in A$, $s \in S \subseteq T$), viewing $a, s \in B$ (as $A \subseteq B$), $s \in T$ directly gives $a/s \in T^{-1}B$. ∎

Combining the previous two results, for a directed union of subrings rather than a single fixed ring:

**Lemma 3.6.10 (Localisation Commutes with Directed Unions of Subrings)**
*Let $(A_i)_{i\in I}$ be a directed system of subrings of a common ring $C$, with union $A := \bigcup_i A_i$, and $T \subset A$ a multiplicative set. Writing $T_i := T \cap A_i$, and realising all localisations as subrings of $C$ (or of a common ring of fractions),*

$$T^{-1}A = \bigcup_i T_i^{-1}A_i$$

*In particular, for $\mathfrak{p} \lhd A$ a prime ideal and $\mathfrak{p}_i := \mathfrak{p} \cap A_i$, taking $T := A \setminus \mathfrak{p}$ (so $T_i = A_i \setminus \mathfrak{p}_i$) gives $A_\mathfrak{p} = \bigcup_i (A_i)_{\mathfrak{p}_i}$.*

*Proof.* If $a/s \in T^{-1}A$ ($a \in A$, $s \in T$), then $a, s \in A_i$ for some $i$ (directed union), with $s \in T \cap A_i = T_i$, so $a/s \in T_i^{-1}A_i$. Conversely if $a/s \in T_i^{-1}A_i$ ($a, s \in A_i$, $s \in T_i \subseteq T$), then viewing $a, s \in A_i \subset A$, $s \in T$, so $a/s \in T^{-1}A$. ∎

## 3.6.2 Modules

**Definition 3.6.11 (Localisation of a module)**
*Let $A$ be a ring with $S$ multiplicative set and $M$ an $A$-module. Then we define*

$$S^{-1}M = \left\{ \frac{m}{s} \;\middle|\; m \in M \right\}$$

*under the obvious equivalence relation. This is then an $S^{-1}A$-module in the obvious way.*

**Definition 3.6.12 (Localisation of a sub-module)**
*Let $M$ be an $A$-module and $N \subseteq M$ a sub-$A$-module then define*

$$S^{-1}N = \left\{ \frac{n}{s} \;\middle|\; n \in M\ s \in S \right\} \subseteq S^{-1}M$$

**Proposition 3.6.13**
*$S^{-1}(-)$ constitutes a functor $A$-**Mod** $\to S^{-1}A$-**Mod**. More precisely there is a unique morphism $\psi$ making the following diagram commute as $A$-module morphisms*

$$\begin{array}{ccc} N & \xrightarrow{\psi} & M \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_S} \\ S^{-1}N & \xrightarrow{S^{-1}(\psi)} & S^{-1}M \end{array}$$

*where $S^{-1}(\psi)$ is in fact an $S^{-1}A$-module morphism.*

*It is an exact functor; for an exact sequence*

$$N \to M \to P$$

*the corresponding sequence of $S^{-1}A$-module morphisms*

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$$S^{-1}N \to S^{-1}M \to S^{-1}P$$

*is exact. If $N$ is a submodule of $M$ then we may regard $S^{-1}N$ as a submodule of $S^{-1}M$.*

**Proposition 3.6.14 (Localisation commutes with quotients)**
*There is a commutative diagram of $A$-module morphisms*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N & \xrightarrow{i} & M & \xrightarrow{\pi} & M/N & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_S} & & \downarrow & & \\ 0 & \longrightarrow & S^{-1}N & \longrightarrow & S^{-1}M & \xrightarrow{S^{-1}(\pi)} & S^{-1}(M/N) & \longrightarrow & 0 \end{array}$$

*with exact rows and the bottom row consists of $S^{-1}A$-module morphisms. This induces an isomorphism of $S^{-1}A$-modules.*

$$S^{-1}M/S^{-1}N \cong S^{-1}(M/N)$$

**Proposition 3.6.15**
*Suppose $N \subseteq N' \subseteq M$ then there is a canonical short-exact sequence of $S^{-1}A$-modules*

$$0 \to S^{-1}(N'/N) \to S^{-1}(M/N) \to S^{-1}(M/N') \to 0$$

*which induces an isomorphism*

$$S^{-1}(M/N)/S^{-1}(N'/N) \cong S^{-1}(M/N')$$

**Proposition 3.6.16**
*Let $M$ be a finitely-generated $A$-module. Then*

$$S^{-1}M = 0 \iff sM = 0 \text{ some } s \in S$$

### 3.6.3 Ideals

Recall the notion of extended and contracted ideals in Definition (3.3.48).

**Definition 3.6.17 (Localisation of an ideal)**
*Let $A$ be a ring and $S$ a multiplicative set and $\mathfrak{a} \lhd A$ define*

$$S^{-1}\mathfrak{a} := \left\{ \frac{a}{s} \;\middle|\; a \in \mathfrak{a} \right\}$$

*then this is an ideal of $S^{-1}A$.*

**Proposition 3.6.18 (Extension and Contraction)**
*Let $A$ be a ring with multiplicative set $S$ and canonical morphism $i_S : A \to S^{-1}A$.*

*a)* $\mathfrak{a}^e = i_S(\mathfrak{a})S^{-1}A = \left\{ \frac{a}{s} \mid a \in \mathfrak{a}, s \in S \right\} = S^{-1}\mathfrak{a}$

*b)* $\mathfrak{b}^c = \left\{ a \mid \frac{a}{1} \in \mathfrak{b} \right\}$

*c) An ideal $\mathfrak{a}$ in $A$ satisfies*

$$\mathfrak{a}^{ec} = \bigcup_{s \in S} (\mathfrak{a} : s) = \{ a \in A \mid as \in \mathfrak{a} \text{ some } s \in S \}$$

*In particular $\mathfrak{a}$ is contracted if and only if $as \in \mathfrak{a} \wedge s \in S \implies a \in \mathfrak{a}$*

*d)* $\mathfrak{b}$ proper $\iff \mathfrak{b}^c$ proper $\iff \mathfrak{b}^c \cap S = \varnothing$

*e)* $\mathfrak{a}^e$ proper $\iff \mathfrak{a} \cap S = \varnothing$

*f) Every ideal $\mathfrak{b} \lhd S^{-1}A$ is extended (equiv. $\mathfrak{b} = \mathfrak{b}^{ce} = S^{-1}\mathfrak{b}^c$).*

*g) A prime ideal $\mathfrak{p}$ is contracted if and only if $\mathfrak{p} \cap S = \varnothing$. In this case $\mathfrak{p}^e$ is prime.*

*h)* $\mathfrak{q}$ prime $\implies \mathfrak{q}^c$ *is prime and satisfies* $\mathfrak{q}^c \cap S = \varnothing$.

*Proof.*

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a) $S^{-1}\mathfrak{a}$ is an additive subgroup because $\frac{a_1}{s_1}+\frac{a_2}{s_2}=\frac{a_1s_2+a_2s_1}{s_1s_2}$. It contains $i_S(\mathfrak{a})$ and is closed under multiplication by $A$, therefore $\mathfrak{a}^e \subseteq S^{-1}\mathfrak{a}$. Similarly as $\mathfrak{a}^e$ is an ideal containing $i_S(\mathfrak{a})$, we have $\frac{a}{s}=\frac{1}{s}\cdot\frac{a}{1}\in\mathfrak{a}^e$, i.e. $S^{-1}\mathfrak{a}\subseteq\mathfrak{a}^e$ as required.

b) This is clear

c) Observe that

$$\begin{aligned}\mathfrak{a}^{\mathrm{ec}} &= \left\{a\in A \;\middle|\; \frac{a}{1}\in\mathfrak{a}^e\right\}\\ &= \left\{a\in A \;\middle|\; \frac{a}{1}=\frac{a'}{s} \quad a'\in\mathfrak{a}\, s\in S\right\}\\ &= \{a\in A \mid sa\in\mathfrak{a} \text{ some } s\in S\}\end{aligned}$$

By (3.3.49) an ideal $\mathfrak{a}$ is contracted if and only if $\mathfrak{a}=\mathfrak{a}^{\mathrm{ec}}$. Furthermore it always satisfies $\mathfrak{a}^{\mathrm{ec}}\subseteq\mathfrak{a}$. The reverse inclusion is precisely the condition given.

d) This first equivalence is true in general, see (3.3.49). Clearly $\mathfrak{b}^c=A\implies\mathfrak{b}^c\cap S\neq\varnothing$. Similarly if $S\cap\mathfrak{b}^c\neq\varnothing$ then $s\in\mathfrak{b}^c\implies\frac{s}{1}\in\mathfrak{b}\implies 1\in\mathfrak{b}\implies 1\in\mathfrak{b}^c$.

e) By d) $\mathfrak{a}^e$ is proper if and only if $\mathfrak{a}^{\mathrm{ec}}$ is proper. By c) we see $1\in\mathfrak{a}^{\mathrm{ec}}$ if and only if $S\cap\mathfrak{a}\neq\varnothing$ and the result follows.

f) By (3.3.49) we need only show $\mathfrak{b}^{\mathrm{ce}}\subseteq\mathfrak{b}$. Note $\frac{a}{s}\in\mathfrak{b}^{\mathrm{ce}}\implies\frac{a}{s}=\frac{a'}{s'}$ with $a'\in\mathfrak{b}^c$. By 2. $\frac{a'}{1}\in\mathfrak{b}$ and therefore so is $\frac{a}{s}=\frac{a'}{s'}=\frac{a'}{1}\cdot\frac{1}{s'}\in\mathfrak{b}$ as required.

g) If $\mathfrak{p}\cap S=\varnothing$ then by primality it automatically satisfies the conditions in c) and is therefore contracted. Conversely if a prime ideal $\mathfrak{p}$ is contracted then $\mathfrak{p}=\mathfrak{q}^c$. It is by definition proper so by d) it satisfies $\mathfrak{p}\cap S=\varnothing$ as required.

Suppose $\mathfrak{p}$ is prime and $\frac{a}{s}\cdot\frac{b}{t}\in\mathfrak{p}^e$ then $\frac{ab}{st}=\frac{x}{u}$ for $x\in\mathfrak{p}\implies v(abu-xst)=0\implies uvab\in\mathfrak{p}\implies a\in\mathfrak{p}$ or $b\in\mathfrak{p}$. Therefore $\frac{a}{s}\in\mathfrak{p}^e$ or $\frac{b}{t}\in\mathfrak{p}^e$ as required.

h) Generically $\mathfrak{q}^c$ is a contracted prime ideal and we've already shown in d) that $\mathfrak{q}^c\cap S=\varnothing$.

■

**Corollary 3.6.19 (Ideal Structure Localisation)**
*Let $A$ be a ring and $S$ a multiplicative set then there is an order-preserving bijection of proper ideals*

$$\{\mathfrak{a}\lhd A \mid \mathfrak{a} \text{ contracted}\}\leftrightarrow\{\mathfrak{b}\lhd S^{-1}A\}$$

*which restricts to a bijection of prime ideals*

$$\{\mathfrak{p}\lhd A \mid \mathfrak{p}\cap S=\varnothing\}\leftrightarrow\{\mathfrak{q}\lhd S^{-1}A\}$$

*Proof.* From (3.6.18) every ideal of $S^{-1}A$ is extended. Therefore the bijection of proper ideals follows from (3.3.52). For prime ideals each direction is well-defined by (3.6.18) g). ■

## 3.6.4 Change of Rings

For what follows it is useful to have the concept of saturation of a multiplicatively closed set. Essentially taking the saturation $\overline{S}$ of $S$ doesn't change the ring $S^{-1}A$.

**Proposition 3.6.20 (Saturation)**
*Let $A$ be a ring and $S$ a multiplicatively closed set. Then the following sets are equal*

*a)* $(i_S)^{-1}\left((S^{-1}A)^\star\right)$
*b)* $\{x\in A \mid ax\in S \text{ for some } a\in A\}$
*c)* $\bigcap_{T\supseteq S:T \text{ saturated}} T$

*which we denote by $\overline{S}$. We have the following properties*

- *$\overline{S}$ is saturated.*
- *$S$ is saturated if and only if $S=\overline{S}$*
- *$\overline{\overline{S}}=\overline{S}$.*

*Proof.* Note $x\in(i_S)^{-1}\left((S^{-1}A)^\star\right)\implies\frac{x}{1}\cdot\frac{b}{t}=1\implies s(xb-t)=0\implies(sb)x\in S$. Similarly if $ax\in S$ then $\frac{x}{1}\cdot\frac{a}{ax}=1$.

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It's clear from b) that the set thus defined is saturated and multiplicatively closed. Let $T$ be another saturated multiplicatively closed set containing $S$ and suppose $ax \in S \implies ax \in T \implies x \in T$, so we find that the sets are equal.

We've proved that $\overline{S}$ is saturated. Clearly $S = \overline{S}$ implies $S$ is saturated. Conversely if $S$ is saturated then by c) we have $\overline{S} \subseteq S$, and clearly $S \subseteq \overline{S}$. The final part follows easily. ∎

We also give another characterization of saturated multiplicatively closed subsets

**Proposition 3.6.21**
*Let $A$ be a ring and $S$ a multiplicatively closed subset. Then*

$$\overline{S} = A \setminus \bigcup_{\mathfrak{p} \cap S = \varnothing} \mathfrak{p}$$

*Proof.* Denote the right hand side by $T$. Then clearly $S \subseteq T$ and as noted before in (3.6.2) $T$ is saturated. Therefore $\overline{S} \subseteq T$ by (3.6.20) c).

Conversely suppose $a \notin \overline{S}$. Consider the principal ideal $(a)$ then $(a) \cap S = \varnothing$ (because $ab \in S \implies a \in \overline{S}$ by (3.6.20) b)). Therefore by (3.3.40) there is a prime ideal $\mathfrak{p}$ containing $a$ which does not intersect $S$. Therefore $a \notin T$. Contrapositively $T \subseteq \overline{S}$ as required. ∎

**Proposition 3.6.22 (Change of Rings)**
*Let $\phi : A \to B$ be a ring homomorphism, $S, T$ corresponding multiplicative subsets. Then*

- *There exists a morphism $\tilde{\phi}$ making the diagram commute*

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_T} \\ S^{-1}A & \overset{\tilde{\phi}}{\dashrightarrow} & T^{-1}B \end{array}$$

  *if and only if $\phi(S) \subseteq \overline{T}$. In this case it is unique and given by*

$$\tilde{\phi}\left(\frac{a}{s}\right) = \frac{\phi(a)b'}{\phi(s)b'}$$

  *where $b' \in B$ is any $b'$ such that $\phi(s)b' \in T$.*

- *If in addition $T \subseteq \phi\left(\overline{S}\right)$ then $\phi$ injective (resp. surjective, bijective) implies $\tilde{\phi}$ is injective (resp. surjective, bijective)*

- *Further $\phi$ surjective $\implies \ker\left(\tilde{\phi}\right) = S^{-1}\ker(\phi)$.*

*Proof.*

- If $\tilde{\phi}$ is well-defined, then $i_T(\phi(S)) = \tilde{\phi}(i_S(S)) \subseteq \tilde{\phi}\left(\left(S^{-1}A\right)^\star\right) \subseteq \left(T^{-1}B\right)^\star$, which implies $\phi(S) \subseteq i_T^{-1}\left(\left(T^{-1}B\right)^\star\right) = \overline{T}$.

  Conversely if $\phi(S) \subseteq \overline{T}$ then $(i_T \circ \phi)(S) \subseteq \left(T^{-1}B\right)^\star$ therefore by (3.6.5) the morphism exists making the diagram commute.

  Note that

$$\tilde{\phi}\left(\frac{a}{s}\right) = \tilde{\phi}(i_S(a)i_S(s)^{-1}) = \tilde{\phi}(i_S(a))\tilde{\phi}(i_S(s))^{-1}$$

  so it is uniquely defined by the commutativity condition. Note that given $s \in S$, by (3.6.20) there exists $b' \in B$ such that $\phi(s)b' \in T$. In this case it's clear that $i_T(\phi(s))^{-1} = \frac{b'}{\phi(s)b'}$ from which the explicit form results.

- Suppose $T \subseteq \phi\left(\overline{S}\right)$ and $\phi$ is injective. Then $\tilde{\phi}\left(\frac{a}{s}\right) = 0 \implies t\phi(a) = 0$ for $t \in T$. Then there exists $s' \in \overline{S}$ and $x \in A$ such that $xs' \in S$ and $\phi(s') = t$. Therefore $\phi(as') = 0 \implies as' = 0 \implies a(xs') = 0 \implies \frac{a}{s} = 0$ as required.

  Similarly if $\phi$ is surjective and given $\frac{b}{t} \in T^{-1}B$ there exists $a \in A$ such that $\phi(a) = b$ and $s \in \overline{S}$ such that $\phi(s) = t$. Then $xs \in S$, $\phi(xs) \in \overline{T}$ and $\phi(yxs) \in T$ for some $x, y \in A$. Finally

$$\tilde{\phi}\left(\frac{axy}{sxy}\right) = \frac{\phi(axy)}{\phi(sxy)} = \frac{b}{t}$$

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as required.

- TODO

■

**Corollary 3.6.23**
*Let $A \xrightarrow{\phi} B \xrightarrow{\psi} C$ be a sequence of homomorphisms and $S, T, U$ be multiplicative sets such that $\phi(S) \subseteq \overline{T}$ and $\psi(T) \subseteq \overline{U}$, then in the notation of the previous Proposition*

$$\tilde{\psi} \circ \tilde{\phi} = \widetilde{\psi \circ \phi}$$

*Proof.* This follows from the uniqueness condition in (3.6.22). ■

**Corollary 3.6.24 (Localisation Maps)**
*Let $A$ be a ring and $S, T$ two multiplicative sets. Then TFAE*

- *There exists $i_{ST} : S^{-1}A \to T^{-1}A$ such that $i_{ST} \circ i_S = i_T$*
- *$S \subseteq \overline{T}$*

*In this case $i_{ST}$ is the unique such map. We have the transitivity relationships*

$$i_{TU} \circ i_{ST} = i_{SU}$$

$$i_{SS} = \mathbf{1}_{S^{-1}A}$$

*and furthermore $i_{ST}$ is an isomorphism if and only if $\overline{S} = \overline{T}$. In particular $i_{S\overline{S}}$ is an isomorphism.*

*Finally* $\ker(i_{ST}) = S^{-1}\ker(i_T)$.

*Proof.* This existence of $i_{ST}$ follows from (3.6.22) when considering the map $\phi = 1_A$. The transitivity and reflexive relationships follow from (3.6.23). ■

**Corollary 3.6.25 (Localisation commutes with quotient)**
*Let $A$ be a ring, $\mathfrak{a}$ an ideal and $S$ a multiplicative set. Then there exists a unique morphism making the diagram commute*

$$\begin{array}{ccc} A & \xrightarrow{\pi} & A/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_{\pi(S)}} \\ S^{-1}A & & \\ \downarrow{\scriptstyle \pi} & & \\ S^{-1}A/S^{-1}\mathfrak{a} & \xrightarrow{\sim} & \pi(S)^{-1}(A/\mathfrak{a}) \end{array}$$

*which is an isomorphism, and determined by*

$$\frac{a}{s} + S^{-1}\mathfrak{a} \to \frac{a+\mathfrak{a}}{s+\mathfrak{a}}$$

*Note that $S \cap \mathfrak{a} \neq \varnothing \iff S^{-1}A/S^{-1}\mathfrak{a} = 0 \iff \pi(S)^{-1}(A/\mathfrak{a}) = 0$.*

*When $\mathfrak{b} \supseteq \mathfrak{a}$ this restricts to a commutative diagram of $A$-modules*

$$\begin{array}{ccc} \mathfrak{b} & \xrightarrow{\pi} & \mathfrak{b}/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_{\pi(S)}} \\ S^{-1}\mathfrak{b} & & \\ \downarrow{\scriptstyle \pi} & & \\ S^{-1}\mathfrak{b}/S^{-1}\mathfrak{a} & \xrightarrow{\sim} & \pi(S)^{-1}(\mathfrak{b}/\mathfrak{a}) \end{array}$$

*and the bottom arrow is still an isomorphism of $S^{-1}A/S^{-1}\mathfrak{a}$-modules.*

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**Corollary 3.6.26 (Localisation commutes with quotient II)**
*Let $A$ be a ring, $\mathfrak{a}$ an ideal and $S$ a multiplicative set. Then there exists a unique morphism making the diagram commute*

$$\begin{array}{ccc} A & \xrightarrow{\pi} & A/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \vdots \\ S^{-1}A & \xrightarrow{\pi} & S^{-1}A/S^{-1}\mathfrak{a} \end{array}$$

*given by*

$$a + \mathfrak{a} \to \frac{a}{1} + S^{-1}\mathfrak{a}$$

*and it is an isomorphism precisely when every $s \in S$ is co-prime to $\mathfrak{a}$, i.e.*

$$(s) + \mathfrak{a} = A \quad \forall s \in S.$$

*When $\mathfrak{b} \supseteq \mathfrak{a}$ this restricts to a commutative diagram*

$$\begin{array}{ccc} \mathfrak{b} & \xrightarrow{\pi} & \mathfrak{b}/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \vdots \\ S^{-1}\mathfrak{b} & \xrightarrow{\pi} & S^{-1}\mathfrak{b}/S^{-1}\mathfrak{a} \end{array}$$

*which is an $A/\mathfrak{a}$-module morphism, and is an isomorphism when the condition (...) holds.*

*Proof.* The condition given for being an isomorphism is equivalent to $\pi(S) \subset (A/\mathfrak{a})^\star$, which is equivalent to $i_{\pi(S)}$ being an isomorphism. Therefore we may apply (3.6.25). ∎

**Corollary 3.6.27**
*Let $A$ be a ring, $\mathfrak{m}$ a maximal ideal and $S$ a multiplicative set such that $S \cap \mathfrak{m} = \varnothing$. Then there is an isomorphism*

$$A/\mathfrak{m} \to S^{-1}A/S^{-1}\mathfrak{m}$$

*In particular $S^{-1}\mathfrak{m}$ is a maximal ideal of $S^{-1}A$.*

**Proposition 3.6.28 (Transitivity)**
*Let $S \subset T$ be multiplicative subsets of $A$ and let*

$$i_S : A \to S^{-1}A$$

*be the localisation at $S$. Define $T_S := i_S(T)$. Then $T_S$ is multiplicative and there is a canonical isomorphism*

$$T^{-1}A \to (T_S)^{-1}(S^{-1}A) \to \left(\overline{T_S}\right)^{-1}(S^{-1}A)$$

*Furthermore if $T \subseteq U$ then $T_S \subseteq U_S$ there is a commutative diagram*

$$\begin{array}{ccccc} & & S^{-1}A & & \\ & \swarrow^{i_{ST}} & \downarrow{\scriptstyle i_{T_S}} & \searrow^{i_{\overline{T_S}}} & \\ T^{-1}A & \xrightarrow{\sim} & (T_S)^{-1}(S^{-1}A) & \xrightarrow{\sim} & \left(\overline{T_S}\right)^{-1}(S^{-1}A) \\ \downarrow{\scriptstyle i_{TU}} & & \downarrow & & \downarrow \\ U^{-1}A & \xrightarrow{\sim} & (U_S)^{-1}(S^{-1}A) & \xrightarrow{\sim} & \left(\overline{U_S}\right)^{-1}(S^{-1}A) \end{array}$$

## 3.6.5 Algebras

**Lemma 3.6.29 (Localisation is a Base Change)**
*Let $A$ be a ring, $S$ a multiplicative subset, and $M$ an $A$-module (3.6.11). Then there is a natural isomorphism of $S^{-1}A$-modules*

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$$S^{-1}A \otimes_A M \xrightarrow{\sim} S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s}.$$

*Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.*

The map $\Phi : S^{-1}A \times M \to S^{-1}M$, $\Phi\left(\frac{a}{s}, m\right) := \frac{am}{s}$, is well-defined: if $\frac{a}{s} = \frac{a'}{s'}$ in $S^{-1}A$ then $u(as' - a's) = 0$ in $A$ for some $u \in S$, hence $u(as' - a's)m = 0$ in $M$, i.e. $u(s'(am) - s(a'm)) = 0$, i.e. $\frac{am}{s} = \frac{a'm}{s'}$ in $S^{-1}M$. $\Phi$ is clearly bi-additive, and $A$-balanced since $\Phi\left(\frac{a}{s} \cdot b, m\right) = \frac{abm}{s} = \Phi\left(\frac{a}{s}, bm\right)$ for $b \in A$; so by the universal property of the tensor product (3.4.4) it induces a group homomorphism

$$\phi : S^{-1}A \otimes_A M \to S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s},$$

which is visibly $S^{-1}A$-linear.

Conversely define $\psi : S^{-1}M \to S^{-1}A \otimes_A M$ by $\psi\left(\frac{m}{s}\right) := \frac{1}{s} \otimes m$. This is well-defined: if $\frac{m}{s} = \frac{m'}{s'}$ in $S^{-1}M$ then $u(s'm - sm') = 0$ for some $u \in S$, and then, using $\frac{1}{s} \otimes m = \frac{us'}{uss'} \otimes m = \frac{1}{uss'} \otimes us'm$ (moving $us' \in A$ across the tensor),

$$\frac{1}{s} \otimes m - \frac{1}{s'} \otimes m' = \frac{1}{uss'} \otimes us'm - \frac{1}{uss'} \otimes usm' = \frac{1}{uss'} \otimes u(s'm - sm') = 0.$$

Then $\phi\left(\psi\left(\frac{m}{s}\right)\right) = \phi\left(\frac{1}{s} \otimes m\right) = \frac{m}{s}$, so $\phi \circ \psi = \mathbf{1}$. Also $\psi\left(\phi\left(\frac{a}{s} \otimes m\right)\right) = \psi\left(\frac{am}{s}\right) = \frac{1}{s} \otimes am = \frac{a}{s} \otimes m$ (again moving $a$ across the tensor), and since elementary tensors generate $S^{-1}A \otimes_A M$, $\psi \circ \phi = \mathbf{1}$. Hence $\phi$ is an isomorphism, with inverse $\psi$. ∎

**Corollary 3.6.30 (Localisation of an Algebra is a Base Change)**
*Let $A$ be a ring, $S$ a multiplicative subset, and $B$ an $A$-algebra. Then $S^{-1}B$ is naturally an $S^{-1}A$-algebra (via $S^{-1}A \to S^{-1}B$, $\frac{a}{s} \mapsto \frac{ab_0}{s}$ where $b_0 \in B$ is the image of $1_A$, the unique extension of $A \to B \to S^{-1}B$ along $i_S$ (3.6.5), as $i_S(S) \subset (S^{-1}B)^\star$), and the isomorphism of $S^{-1}A$-modules furnished by (3.6.29) (with $M := B$),*

$$S^{-1}A \otimes_A B \xrightarrow{\sim} S^{-1}B, \quad \frac{a}{s} \otimes b \mapsto \frac{ab}{s},$$

*is moreover an isomorphism of rings, where $S^{-1}A \otimes_A B$ carries the ring structure of (3.4.34).*

*Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.*

Write $\phi$ for the $S^{-1}A$-module isomorphism of (3.6.29). It remains only to check $\phi$ is multiplicative on elementary tensors, since it is already additive:

$$\phi\left(\left(\frac{a}{s} \otimes b\right) \cdot \left(\frac{a'}{s'} \otimes b'\right)\right) = \phi\left(\frac{aa'}{ss'} \otimes bb'\right) = \frac{aa'bb'}{ss'} = \frac{a}{s} \cdot b \cdot \frac{a'}{s'} \cdot b' = \phi\left(\frac{a}{s} \otimes b\right) \cdot \phi\left(\frac{a'}{s'} \otimes b'\right).$$

As every element of $S^{-1}A \otimes_A B$ is a sum of elementary tensors, and $\phi$ is additive, $\phi$ is a ring homomorphism; being also a bijection, it is a ring isomorphism. ∎

**Corollary 3.6.31 (Localisation of a Domain is a Base Change to the Fraction Field)**
*Let $A \subseteq B$ be an inclusion of integral domains, $K := \mathrm{Frac}(A)$, and $S := A \setminus \{0\}$, so that $S^{-1}B$ is naturally a subring of $\mathrm{Frac}(B)$ (as $S \subset B \setminus \{0\}$ too). Then there is a natural isomorphism of rings*

$$B \otimes_A K \xrightarrow{\sim} S^{-1}B, \quad b \otimes x \mapsto bx,$$

*where the right-hand side is computed in $\mathrm{Frac}(B)$, regarding $B, K \subseteq \mathrm{Frac}(B)$.*

*Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.*

By (3.6.30) (applied to the $A$-algebra $B$, noting $S^{-1}A = K$) there is a ring isomorphism $K \otimes_A B \xrightarrow{\sim} S^{-1}B$, $\frac{a}{s} \otimes b \mapsto \frac{ab}{s}$. Composing with the symmetry isomorphism $B \otimes_A K \cong K \otimes_A B$, $b \otimes x \mapsto x \otimes b$ (3.4.7) (which is also a ring isomorphism, since swapping factors clearly preserves the componentwise multiplication of (3.4.34)), yields a ring isomorphism $B \otimes_A K \xrightarrow{\sim} S^{-1}B$ sending $b \otimes \frac{a}{s} \mapsto \frac{ab}{s}$; taking $a = 1$ this is $b \otimes x \mapsto x \cdot b = bx$ for $x = \frac{1}{s}$, and since $x = \frac{a}{s} = a \cdot \frac{1}{s}$ for general $x \in K$, bilinearity gives $b \otimes x = ab \otimes \frac{1}{s} \mapsto \frac{ab}{s} = (ab) \cdot \frac{1}{s} = b \cdot \left(a \cdot \frac{1}{s}\right) = bx$ on every elementary tensor.

Finally $\frac{ab}{s} \in S^{-1}B \subseteq \mathrm{Frac}(B)$, computed there, agrees with the product $bx$ computed in $\mathrm{Frac}(B)$ (as $B, K \subseteq \mathrm{Frac}(B)$). ∎

**Remark 3.6.32**
*(3.6.30) recovers (3.6.41) (relative localisation at a prime) as the special case $S := A \setminus \mathfrak{p}$ of a general ring homomor-*

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*phism $\phi : A \to B$: the module localisation $S^{-1}B$ (with $B$ regarded as an $A$-module via $\phi$) is exactly $\phi(S)^{-1}B = B_{\mathfrak{p}}$, since the two defining equivalence relations on fractions agree termwise.*

### 3.6.6 Localisation at an element

**Definition 3.6.33 (Localisation at an element)**
*Let $A$ be a ring and $f \in A$. Then define*

$$S_f = \{1, f, ..., f^n, ...\}$$

*and*

$$A_f := (S_f)^{-1}A$$

*We have canonical maps*

$$i_f : A \to A_f$$

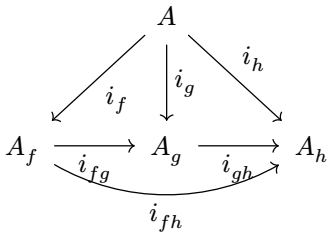
*given by $i_f := i_{S_f}$.*

*Note that $A_f$ is a zero ring iff $f$ is nilpotent.*

**Proposition 3.6.34 (Transition maps for localisation at an element)**
*Let $A$ be a ring and $f, g \in A$ then*

$$S_f \subseteq \overline{S_g} \iff f \mid g^N \text{ some N} > 0$$

*in this case define $i_{fg} = i_{S_f S_g}$ to be the unique morphism such that $i_{fg} \circ i_f = i_g$ (3.6.24). In addition if $h \in A$ and $S_g \subseteq \overline{S_h}$ we have a commutative diagram*




*Furthermore $\overline{S_1} = A^\star$ and $i_1$ is an isomorphism.*

**Proposition 3.6.35 (Transitivity of localizing at elements)**
*Let $A$ be a ring and $f, g \in A$ such that $S_f \subseteq \overline{S_g}$. There is a canonical isomorphism*

$$A_g \xrightarrow{\sim} (A_f)_{g/1}$$

*Furthermore $S_g \subseteq \overline{S_h} \implies S_{g/1} \subseteq \overline{S_{h/1}}$ and there is a commutative diagram*

$$\begin{array}{ccc}
(A_f)_1 & \xrightarrow{\sim} & A_f \\
\downarrow{\scriptstyle i_{1(g/1)}} & & \downarrow{\scriptstyle i_{g/1}} \\
(A_f)_{g/1} & \xrightarrow{\sim} & A_g \\
\downarrow & & \downarrow{\scriptstyle i_{gh}} \\
(A_f)_{h/1} & \xrightarrow{\sim} & A_h
\end{array}$$

*with the horizontal arrows isomorphisms and the vertical arrows are well-defined.*

*Proof.* The existence of the isomorphism is from (3.6.28) as $i_f(S_g) = S_{g/1}$. The second statement follows because $g \mid h^N \implies g/1 \mid h^N/1$ trivially. ∎

**Proposition 3.6.36**
*Let $A$ be a ring and $f \in A$. Then localisation map $A \to A_f$ induces an order-preserving correspondence of prime ideals*

$$\{\mathfrak{p} \lhd A \mid f \notin \mathfrak{p}\} \leftrightarrow \{\mathfrak{p} \lhd A_f\}$$

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*Proof.* The correspondence of prime ideals follows from (3.6.19) once we observe that $S_f \cap \mathfrak{p} = \varnothing \iff f \notin \mathfrak{p}$. ∎

**Proposition 3.6.37**
*Let $A$ be a ring, $S$ a multiplicatively closed subset and $g \in S$. Then the canonical map $A_g \to S^{-1}A$ has kernel $\ker(i_S)_g$. In particular when $\ker(i_S)$ is finitely generated it is injective iff $g^r \ker(i_S) = 0$ for some $r > 0$.*

*When $A$ is Noetherian then there always exists $g$ such $A_g \to S^{-1}A$ is injective, and any multiple of $g$ also satisfies this property.*

*Proof.* The first statement follows from (3.6.4).

When $A$ is Noetherian then $\ker(i_S) = (a_1, ..., a_n)$ for some $a_i \in A$. By definition there exists some $s_i$ such that $s_i a_i = 0$, whence we may take $g = s_1 \cdots s_n$. ∎

**Proposition 3.6.38**
*Let $A$ be a ring, $S$ a multiplicatively closed subset and $M_1, ..., M_n$ a family of $A$-modules. Then there is an isomorphism*

$$S^{-1}\left(\bigoplus_{i=1}^{n} M_i\right) \xrightarrow{\sim} \bigoplus_{i=1}^{n} S^{-1}M_i$$

## 3.6.7 Localisation at a prime ideal

**Definition 3.6.39 (Localisation at a prime ideal)**
*Let $A$ be a ring and $\mathfrak{p} \lhd A$ a prime ideal. Then $S := A \setminus \mathfrak{p}$ is a saturated multiplicatively closed subset, and we define*

$$A_{\mathfrak{p}} := (A \setminus \mathfrak{p})^{-1}A$$

*For an ideal $\mathfrak{a} \lhd A$ write the extended ideal*

$$\mathfrak{a}A_{\mathfrak{p}} := S^{-1}\mathfrak{a} = \mathfrak{a}^e.$$

**Definition 3.6.40 (Relative localisation at a prime ideal)**
*Let $\phi : A \to B$ be a ring homomorphism and $\mathfrak{p} \lhd A$ a prime ideal. Define*

$$B_{\mathfrak{p}} := \phi(A \setminus \mathfrak{p})^{-1}B$$

*For an ideal $\mathfrak{a} \lhd A$ write*

$$\mathfrak{a}B_{\mathfrak{p}} := (\phi(\mathfrak{a})B)B_{\mathfrak{p}}$$

*Observe $B_{\mathfrak{p}} = 0 \iff \mathfrak{p} \subsetneq \ker(\phi)$, so we would typically assume $\ker(\phi) \subseteq \mathfrak{p}$.*

**Proposition 3.6.41 (Relative Localisation as a Tensor Product)**
*Let $\phi : A \to B$ be a ring homomorphism (so $B$ is an $A$-algebra) and $\mathfrak{p} \lhd A$ a prime ideal, $S := A \setminus \mathfrak{p}$. Then $B_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-algebra via $\frac{a}{s} \mapsto \phi(a)/\phi(s)$ (well-defined as $\phi(S) \subseteq B_{\mathfrak{p}}^{\star}$ already), and there is an isomorphism of $A_{\mathfrak{p}}$-algebras*

$$\Psi : B \otimes_A A_{\mathfrak{p}} \xrightarrow{\sim} B_{\mathfrak{p}}, \quad b \otimes \frac{a}{s} \mapsto \frac{\phi(a)b}{\phi(s)},$$

*with inverse $\Theta(b/\phi(s)) := b \otimes \left(\frac{1}{s}\right)$.*

*Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.*

**$\Psi$ is well-defined.** For fixed $\frac{a}{s} \in A_{\mathfrak{p}}$, $b \mapsto \phi(a)b/\phi(s)$ is $A$-linear in $b$ ($B_{\mathfrak{p}}$ being an $A$-module via $\phi$); for fixed $b \in B$, if $\frac{a}{s} = \frac{a'}{s'}$ in $A_{\mathfrak{p}}$ then $s''(as' - a's) = 0$ for some $s'' \in S$, so applying $\phi$ and multiplying by $b$, $\phi(s'')(\phi(a)\phi(s') - \phi(a')\phi(s))b = 0$ in $B$, giving $\phi(a)b/\phi(s) = \phi(a')b/\phi(s')$ in $B_{\mathfrak{p}}$; and $\frac{a}{s} \mapsto \phi(a)b/\phi(s)$ is $A$-linear in $\frac{a}{s}$ (direct check, as in (3.6.5)). So the map $B \times A_{\mathfrak{p}} \to B_{\mathfrak{p}}$, $\left(b, \frac{a}{s}\right) \mapsto \phi(a)b/\phi(s)$, is $A$-bilinear, and induces $\Psi$ by the universal property of the tensor product (3.4.4).

**$\Theta$ is well-defined.** Write $a \cdot b := \phi(a)b$ for the $A$-module structure on $B$. If $b/\phi(s) = b'/\phi(s')$ in $B_{\mathfrak{p}}$ then $s'' \cdot (s' \cdot b) = s'' \cdot (s \cdot b')$ in $B$ for some $s'' \in S$; using $(a \cdot b) \otimes x = b \otimes (ax)$ (the defining relation of $\otimes_A$) twice,

$$b \otimes \frac{1}{s} = (s''s') \cdot b \otimes \frac{1}{s''s's} = (s'' \cdot (s' \cdot b)) \otimes \frac{1}{s''s's} = (s'' \cdot (s \cdot b')) \otimes \frac{1}{s''ss'} = b' \otimes \frac{1}{s'},$$

so $\Theta$ is well-defined on $B_{\mathfrak{p}}$.

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**$\Theta$ and $\Psi$ are mutually inverse.** $\Psi\Theta(b/\phi(s)) = \Psi\left(b \otimes \frac{1}{s}\right) = \phi(1)b/\phi(s) = b/\phi(s)$. And $\Theta\Psi\left(b \otimes \frac{a}{s}\right) = \Theta(\phi(a)b/\phi(s)) = (\phi(a)b) \otimes \left(\frac{1}{s}\right) = b \otimes \left(\frac{a}{s}\right)$ (the same tensor relation as above), so $\Theta\Psi$ is the identity on simple tensors, hence (by linearity) on all of $B \otimes_A A_\mathfrak{p}$.

**$\Psi$ is $A_\mathfrak{p}$-linear.** For $\frac{a'}{s'} \in A_\mathfrak{p}$,

$$\Psi\left(\frac{a'}{s'} \cdot \left(b \otimes \frac{a}{s}\right)\right) = \Psi\left(b \otimes \frac{aa'}{ss'}\right) = \frac{\phi(aa')b}{\phi(ss')} = \frac{\phi(a')}{\phi(s')} \cdot \frac{\phi(a)b}{\phi(s)} = \frac{a'}{s'} \cdot \Psi\left(b \otimes \frac{a}{s}\right),$$

using the $A_\mathfrak{p}$-algebra structure on $B_\mathfrak{p}$ displayed above; so $\Psi$ is $A_\mathfrak{p}$-linear on simple tensors, hence (by linearity) on all of $B \otimes_A A_\mathfrak{p}$.

**Multiplicativity.** By the algebra tensor product structure on $B \otimes_A A_\mathfrak{p}$, $\left(b \otimes \frac{a}{s}\right)\left(b' \otimes \frac{a'}{s'}\right) = bb' \otimes \frac{aa'}{ss'}$, and

$$\Psi\left(\left(b \otimes \frac{a}{s}\right)\left(b' \otimes \frac{a'}{s'}\right)\right) = \frac{\phi(aa')bb'}{\phi(ss')} = \frac{\phi(a)b}{\phi(s)} \cdot \frac{\phi(a')b'}{\phi(s')} = \Psi\left(b \otimes \frac{a}{s}\right)\Psi\left(b' \otimes \frac{a'}{s'}\right),$$

so $\Psi$ is multiplicative on simple tensors, hence (by linearity) a ring homomorphism – and so, with $A_\mathfrak{p}$-linearity above, an isomorphism of $A_\mathfrak{p}$-algebras. ∎

**Proposition 3.6.42**
*Let $A$ be a ring and $\mathfrak{p}$ a prime ideal. Consider the localisation $A \to A_\mathfrak{p}$. Then there is an order-preserving bijection between (prime) ideals contained in $\mathfrak{p}$ and (prime) ideals of $A_\mathfrak{p}$*

$$\{\mathfrak{q} \lhd A \mid \mathfrak{q} \subseteq \mathfrak{p}\} \leftrightarrow \{\mathfrak{q} \lhd A_\mathfrak{p}\}$$
$$\mathfrak{q} \to \mathfrak{q}A_\mathfrak{p}$$

*In particular $A_\mathfrak{p}$ is a local ring with unique maximal ideal $\mathfrak{p}A_\mathfrak{p}$.*

*Proof.* Clearly $\mathfrak{q} \cap (A \setminus \mathfrak{p}) = \varnothing \iff \mathfrak{q} \subseteq \mathfrak{p}$, so the result follows from (3.6.19). ∎

**Proposition 3.6.43**
*Let $A$ be a ring and $\mathfrak{p}$ a prime ideal. Then the map*

$$\mathrm{Frac}(A/\mathfrak{p}) \xrightarrow{\sim} A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$$
$$\frac{\overline{a}}{\overline{b}} \to \frac{\overline{a/1}}{\overline{b/1}} = \overline{\left(\frac{a}{b}\right)}$$

*is a ring isomorphism*

*Proof.* Consider the ring map $A \to A_\mathfrak{p} \to A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$. Then this has kernel $\mathfrak{p}^{ec} = \mathfrak{p}$ (3.6.18), so may be factored as an injective map $A/\mathfrak{p} \hookrightarrow A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$. By (3.6.42) $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ is a field so by universal property of localisation (3.6.5) this extends to an injective map $\mathrm{Frac}(A/\mathfrak{p}) \hookrightarrow A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ of the given form. We may verify directly that it is surjective. ∎

**Proposition 3.6.44 (Localisation at prime is direct limit of localisation at an element)**
*Let $A$ be a ring and $\mathfrak{p}$ a prime ideal then*

$$S_\mathfrak{p} := A \setminus \mathfrak{p} = \bigcup_{f \in A \setminus \mathfrak{p}} \overline{S_f}$$

*Therefore there are canonical morphisms (for $f \notin \mathfrak{p}$)*

$$i_{S_f S_\mathfrak{p}} : A_f \to A_\mathfrak{p}$$

*Furthermore the family of multiplicatively closed sets $\{S_f\}_{f \notin \mathfrak{p}}$ (resp. rings $\{A_f\}_{f \notin \mathfrak{p}}$) form a directed system under the relation $S \prec T \iff S \subseteq \overline{T}$. Therefore we have a canonical ring homomorphism*

$$\varinjlim_{f \notin \mathfrak{p}} A_f \to A_\mathfrak{p}$$

*which is an isomorphism.*

*Proof.* As $A \setminus \mathfrak{p}$ is a saturated multiplicatively closed set we have $f \in A \setminus \mathfrak{p} \iff S_f \subseteq A \setminus \mathfrak{p} \iff \overline{S_f} \subseteq A \setminus \mathfrak{p}$. Therefore the expression for $S_\mathfrak{p}$ follows.

The family of multiplicatively closed subsets is a directed system because $S_f \subseteq \overline{S_{fg}}$. To see this note $fg \in \overline{S_{fg}} \implies f \in \overline{S_{fg}} \implies S_f \subseteq \overline{S_{fg}}$.

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The final isomorphism follows because we can decompose it into two maps

$$\varinjlim_{f\notin\mathfrak{p}} A_f \xrightarrow{\sim} \varinjlim_{f\notin\mathfrak{p}} \overline{S_f}^{-1}A \xrightarrow{\sim} A_{\mathfrak{p}}$$

The first is an isomorphism by (3.6.24) and the second by (3.6.8), in light of the first statement. ∎

**Proposition 3.6.45 (Quotient commutes with Localisation at Prime ideal)**
*Let $A$ be a ring, $\mathfrak{a}$ an ideal and $\mathfrak{p}\supset\mathfrak{a}$ a prime ideal. Then there is an isomorphism of non-zero rings*

$$A_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}} \cong (A/\mathfrak{a})_{\mathfrak{p}/\mathfrak{a}}$$
$$\frac{a}{s}+\mathfrak{a}_{\mathfrak{p}} \to \frac{a+\mathfrak{a}}{s+\mathfrak{a}}$$

*Proof.* Let $\pi: A\to A/\mathfrak{a}$ be the quotient map then we claim $\pi(A\setminus\mathfrak{p})=\pi(A)\setminus\pi(\mathfrak{p})$. As $\pi$ is surjective we only need to show that $\pi^{-1}(\pi(\mathfrak{p}))=\mathfrak{p}$. That is if $\pi(a)=\pi(b)$ and $b\in\mathfrak{p}$ then $a\in\mathfrak{p}$. However this follows by assumption because $\mathfrak{a}\subset\mathfrak{p}$. Therefore we may apply (3.6.25). ∎

**Proposition 3.6.46 (Minimal Primes consist of Zero Divisors)**
*Let $\mathfrak{p}$ be a proper prime ideal minimal over $\mathfrak{a}$. Then for every $x\in\mathfrak{p}$ there exists $a\in A\setminus\mathfrak{a}$ such that $xa\in\mathfrak{a}$.*

*In particular a minimal prime ideal consists of zero-divisors.*

*Proof.* Observe that $\mathfrak{p}/\mathfrak{a}$ is a proper minimal prime ideal of $A/\mathfrak{a}$ by (3.3.55). Then the first statement is equivalent to showing every $x\in\mathfrak{p}/\mathfrak{a}$ is a zero divisor and we may reduce to the case $\mathfrak{a}=(0)$.

Consider the ring $A_{\mathfrak{p}}$, it has a unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ which is also minimal by (3.6.42). Every other prime ideal is contained in $\mathfrak{p}A_{\mathfrak{p}}$ by (3.3.36), and therefore equal by minimality. Therefore $\sqrt{(0)A_{\mathfrak{p}}}=\mathfrak{p}A_{\mathfrak{p}}$ by (3.3.45). Suppose $x\in\mathfrak{p}$ then $\frac{x^n}{1}=0$ and $tx^n\in\mathfrak{a}$ for $t\notin\mathfrak{p}$. Choose $n$ minimal subject to this condition. Then $n\geq 1$ because $t\notin\mathfrak{a}$, and so $a=tx^{n-1}$ has the required properties. ∎

## 3.6.8 Finiteness Results

**Lemma 3.6.47**
*Let $M$ be an $A$-module, $N$ a submodule and $f_1,...,f_r\in A$ such that*

*a) $N_{f_i}\to M_{f_i}$ is surjective for all $i=1...r$*
*b) $A=(f_1,...,f_r)$ as an ideal*

*then $M=N$.*

*Proof.* Consider the $A$-module $P:=M/N$. Suppose $P\neq 0$, then $0\neq p\in P$ and $\mathrm{Ann}(p)$ is a proper ideal of $A$. In particular $f_i\notin\mathrm{Ann}(p)$ for some $i$. This shows $P_{f_i}\neq 0$. From (3.6.14) we have $P_{f_i}\xrightarrow{\sim} M_{f_i}/N_{f_i}$ and so $N_{f_i}\neq M_{f_i}$ a contradiction. ∎

**Proposition 3.6.48**
*Let $M$ be an $A$-module. Then*

*a) $M$ finitely-generated $\implies M_f$ is a finitely generated $A_f$-module for all $f\in A$*
*b) $(f_1,...,f_r)=A$ and $M_{f_i}$ a finitely generated $A_{f_i}$-module for all $i=1...r\implies M$ is a finitely-generated $A$-module*

*Proof.*
a) Let $m_1,...m_n\in M$ be a generating set. Then it is easy to verify that $m_1/1,...,m_n/1$ is a generating set for $M_f$.
b) For each $i=1...r$, let $m_{ij}$ be such that $m_{ij}/1$ generates $M_{f_i}$ as an $A_{f_i}$-module. Let $N$ be the $A$-submodule of $M$ generated by the $m_{ij}$. Then $N_{f_i}=M_{f_i}$. We conclude by (3.6.47) that $N=M$ is finitely generated as an $A$-module.

∎

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